technology, enabling anyone to participate in the development and deployment of dApps and smart contracts.

Customisability: Developers can modify and customise the codebase of open source blockchain platforms to suit specific use cases and requirements.

Community collaboration*:* Open source projects foster collaboration and knowledge sharing among developers, leading to rapid innovation and iteration.

Interoperability: Many open source blockchain platforms support interoperability, allowing different blockchains to communicate and exchange data seamlessly.

Some of the key benefits of implementing blockchain solutions with open source platforms are:

Cost-effectiveness*:* Open source blockchain platforms are typically free to use, reducing the barrier to entry for developers and organisations.

Security: The transparency of open source platforms enhances security by allowing developers to identify and fix vulnerabilities quickly.

Innovation*:* These platforms foster innovation by enabling collaboration and experimentation among developers from diverse backgrounds and organisations.

Community support: They benefit from a vibrant community of developers, contributors, and enthusiasts who provide support, feedback, and contributions.

Adaptability: Open source platforms are highly adaptable, allowing developers to customise and extend the functionality of the platform to meet evolving needs and requirements.

Private blockchain platforms

Private blockchain platforms are permissioned networks where access is restricted to authorised participants. These platforms offer greater control over governance, privacy, and scalability compared to public blockchains.

Key features of private blockchains include:

Permissioned access: Access to the blockchain network is restricted to approved participants, typically through invitation or authentication mechanisms.

Centralised governance: Governance structures are controlled by designated entities or consortiums, enabling faster decision-making and coordination.

Privacy: Private blockchains offer enhanced privacy features, such as confidential transactions and encrypted data storage, catering to enterprises with strict data privacy requirements.

Scalability: With fewer participants and transaction validators, private blockchains can achieve higher transaction throughput and scalability compared to public blockchains.

Private blockchain platforms can be applied to various industrial use cases where centralised transactions and privacy are the key requirements. These include:

Supply chain management: Private blockchains facilitate supply chain transparency, traceability, and efficiency by securely recording and sharing transaction data among authorised participants.

Table 1: A comparison of the features of various blockchain platforms

Feature	Ethereum	Hyperledger Fabric	EOSIO	Corda
Consensus mechanism	Proof of Work (PoW), transitioning to Proof of Stake (PoS)	Pluggable consensus (e.g., Raft, Kafka, PBFT)	Delegated Proof of Stake (DPoS)	Not applicable (permissioned network)
Smart contracts	Yes	Yes	Yes	Yes
Language support	Solidity, Vyper	Go, Java, JavaScript, TypeScript	C++, WebAssembly (EOSIO C++)	Java, Kotlin
Privacy features	Limited	Private data collections	Limited (focus on performance)	Confidential identities, transactions
Transaction throughput	Variable, but generally lower than EOSIO	Variable, depending on network setup	High	Moderate
Interoperability	Limited	Interoperability via Hyperledger Composer, etc	Limited	Yes (with other Corda networks)
Governance model	Decentralised	Consortium governance	Decentralised	Consortium governance
Community support	High support	Good community support	High support	High support
Use cases	DApps, DeFi, ICOs, NFTs	Enterprise applications, supply chain, healthcare	Enterprise applications, social media, gaming	Financial services, supply chain